

PULSEGUARD TECHNOLOGIES INC. - STANDARD OPERATING PROCEDURE

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Title:	MANDATORY SOFTWARE RELEASE TESTING PROTOCOLS
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Department:	Quality Assurance (QA)
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Approver:	Sneha Deshmukh (VP of Engineering)
Classification:	Internal Restricted

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1. PURPOSE & OBJECTIVE

This Standard Operating Procedure (SOP) defines the mandatory testing protocols and quality checks required before any software release candidate is permitted to deploy to the production environment. The objective is to verify that all features are bug-free, regression suites are executed fully, and compliance parameters are checked, eliminating risks of service interruptions.

2. SCOPE

This procedure is strictly applicable to all release candidate branches within the product lines of PulseGuard Technologies. It covers backend services, frontend interfaces, API endpoints, mobile applications, and internal database migrations. It applies to all QA engineers, developers, and product leads.

3. ROLES AND RESPONSIBILITIES

3.1. QA Lead

- Validates that the testing plan is complete.
- Schedules automation runs and manual regression reviews.
- Issues the final QA Release Sign-off.

3.2. Automation QA Engineer

- Manages and maintains regression test scripts.
- Triggers the CI/CD automation pipeline.
- Reviews and reports test failures.

3.3. Manual QA Tester

- Validates complex edge cases not covered by scripts.
- Audits UI layouts and accessibility compliance.

3.4. Release Manager

- Verifies that QA Sign-off exists prior to prod deployments.
- Blocks deployment if failures or critical defects exist.

4. DEFINITIONS AND REFERENCES

- Regression Testing: A comprehensive suite of tests verifying that existing functionality has not been broken by new changes.
- Automation Coverage: The ratio of automated test cases executing successfully to the total count of required test scenarios.
- Staging: A replica of production used for pre-release verification.
- SOP-QA-002: Test Environment Maintenance (reference document).

5. MANDATORY PRE-RELEASE TESTING CRITERIA

Every release candidate must pass through the following testing criteria:

5.1. Regression Testing Execution

- A full regression test suite must be executed for all release candidates.
- Bypassing the regression suite under release time pressure is forbidden.

5.2. Automation Coverage Threshold

- At least 80% of critical user paths must have automated test scripts passing successfully.
- Major releases require 85% automation passing rate before code sign-off.

5.3. Manual Validation

- The QA team must manually verify all new features and bug fixes on staging before signing off on the release.

5.4. Security Validation

- Vulnerability scanning reports must be reviewed and resolved.

6. DEFECT CLASSIFICATION AND RESOLUTION PROTOCOLS

Defects discovered during testing must be categorized and resolved as follows:

6.1. Critical Severity Defects

- Description: Total crash, security breach, data loss, or blocked flows.
 - Protocol: Deployment is immediately blocked. No exceptions allowed.
- 6.2. Major Severity Defects
- Description: Non-critical feature broken with no easy workaround.
 - Protocol: Deployment is blocked unless signed off by the QA Lead.
- 6.3. Minor Severity Defects
- Description: Typographical errors, minor spacing, visual bugs.
 - Protocol: Deployment may proceed. Defects are logged to Jira backlog.

7. PROCEDURAL STEP-BY-STEP FLOW

- Step 1: Release Candidate (RC) branch is generated by the engineering lead.
- Step 2: RC is auto-deployed to the Staging Environment.
- Step 3: QA Lead triggers the automated regression suite on Staging.
- Step 4: Automation Engineer monitors output and compiles the automation report.
- Step 5: Manual QA Testers begin manual test cases for new feature changes.
- Step 6: Defect reports are logged in the tracking tool.
- Step 7: Developers fix any critical/major defects on staging.
- Step 8: QA Lead verifies the fixes and re-runs the affected test cases.
- Step 9: Once all criteria are met, QA Lead uploads the QA Sign-off Log.
- Step 10: Release Manager initiates production deployment.

8. TROUBLESHOOTING & ESCALATION PROCEDURES

8.1. Regression Test Failures

- If regression tests fail, the build is blocked.
- Bypass requests must be escalated to the QA Director for review.

8.2. Escalation Contact Path

- First Level: Department QA Lead (Diana Patel).
- Second Level: Sneha Deshmukh (VP of Engineering / Escalation Manager).
- Final Approval: Chief Technology Officer.

9. REVISION HISTORY LOG

Version	Date	Author	Description of Change
v1.0	2024-05-14	D. Patel	Initial release of release protocols.
v2.0	2025-02-19	D. Patel	Updated automation coverage to 80%.
v2.8	2026-01-15	D. Patel	Added strict regression bypass ban.